SGPMAIN6.2v.py: User Manual

Updated for Lujo Virus (LUJO) Analysis

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Contents

1 Introduction

1.1 Overview and Motivation

SGPMAIN6.2v.py is the latest version of the Specular Grassmann-Polarity Index Method system. This computational platform for protein analysis moves beyond traditional sequence alignment methods to capture the physicochemical essence of proteins. The central insight is that protein function is fundamentally determined by the spatial and temporal organization of polarity transitions along the polypeptide chain, not by the primary sequence of amino acids per se.

The system is built on three fundamental pillars:

1. **The Polarity Index Method (PIM)**: Transforms amino acid sequences into 16-dimensional vectors representing polarity transition frequencies.
2. **Geometric Algebra Cl(16)**: Provides the mathematical engine for comparing and transforming polarity profiles using true geometric algebra operations.
3. **Derived Chemical Properties and PIDP**: Translates abstract mathematics into actionable biochemical insights.

1.2 What's New in Version 6.2

- **Complete Geometric Algebra Cl(16)**: All grades (0–16) implemented as true operations.
- **True Hodge dual**: With pseudoscalar $I = e_1 \wedge e_2 \wedge \cdots \wedge e_{16}$.
- **True Ricci curvature tensor** in Grassmann space.
- **Complete geometric product**: With all grade combinations.
- **True rotors**: With exponential $\exp(-B/2)$.
- **True Radon transform, fractal dimension, and Moran's I**.
- **Enhanced PIDP Integration**: metapredict and AIUPred with redox sensitivity.
- **Unified Chemical Profiler**: 54 properties in a single CSV file.
- **Therapeutic Peptide Design**: Machine learning-based activity prediction.

1.3 Version Information

This manual corresponds to `SGPMAIN6.2v.py`. The reference analysis processed 151,066,931 sequences across 26 groups in approximately 13 hours and 21 minutes.

2 Installation and System Requirements

2.1 System Requirements

Before installing SGPMAIN6.2v.py, ensure your system meets the following requirements:

Table 1: System requirements

Component	Minimum Requirement
Operating System	Linux, macOS, or Windows (with WSL)
Python	3.8 or higher
RAM	8 GB (16 GB recommended)
Storage	10 GB free space
Processor	64-bit, 2 or more cores

2.2 Required Python Packages

The following Python packages are required:

```
numpy >= 1.21.0
pandas >= 1.3.0
scipy >= 1.7.0
scikit-learn >= 1.0.0
biopython >= 1.79
matplotlib >= 3.4.0
seaborn >= 0.11.0
psutil >= 5.8.0
opencv-python >= 4.5.0
# Optional for PIDP
metapredict >= 1.0
aiupred >= 3.0
```

Listing 1: Required packages

2.3 Installation Steps

1. Clone the repository:

```
git clone https://github.com/polancodf2024/lujv-sgp-analysis.git
cd lujv-sgp-analysis
```

2. Install dependencies:

```
pip install -r requirements.txt
```

3. (Optional) Install PIDP dependencies:

```
pip install metapredict aiupred
```

4. Verify installation:

```
python3 SGPMAIN6.2v.py --help
```

3 Configuration Guide

3.1 Target Group Configuration

The target group is the primary protein or virus family to be analyzed. To analyze Lujo virus:

```
MAIN_GROUP = ['lujo']
```

3.2 Key Configuration Parameters

Table 2: Main configuration parameters

Parameter	Default	Description
MAX_STORED_PROTEINS_PER_GROUP	10000	Maximum proteins stored per group
BATCH_SIZE	100000	Number of sequences per batch
USE_PIDP	True	Enable intrinsic disorder analysis
USE_BIOLOGICAL_METRIC	True	Enable biological metric
USE_KARHUNEN_LOEVE	True	Enable KL decomposition
USE_HODGE_DUAL	True	Enable Hodge dual operations
USE_RICCI_CURVATURE	True	Enable Ricci curvature calculation
TOP_N_PROTEINS	20	Number of top proteins to return

3.3 PIDP Configuration

```
USE_PIDP = True              # False to disable
PIDP_USE_METAPREDICT = True  # Use metapredict
PIDP_USE_AIUPRED = True      # Use AIUPred
PIDP_THRESHOLDS = [0.3, 0.4, 0.5]
```

4 Running the Analysis

4.1 Basic Usage

```
# Run the complete analysis
python3 SGPMAIN6.2v.py

# The results will be saved in:
# results_v17_YYYYMMDD_HHMMSS/
```

4.2 Required Input Files

Table 3: Input files

File	Description
lujo.unico.dat0	Lujo virus sequences
*.unico.dat0	Reference group sequence files
chembl_uniprot.txt	ChEMBL-UniProt mapping (optional)
apd_natural.fasta	Antimicrobial peptide database (optional)

4.3 Example Output

```
[SUCCESS] Comparison saved: results_v17_20260908_.../comparison_lujo_vs_all.csv
[SUCCESS] Similarity matrix saved: similarity_matrix_groups.csv
[SUCCESS] Top 20 individual proteins saved: top_20_individual_proteins.csv
[SUCCESS] Therapeutic profile saved: therapeutic_profile.json
[SUCCESS] Chemical profile saved: chemical_profile_lujo.csv (54 properties)
[SUCCESS] PIDP analysis saved: pidp_analysis_lujo.csv
[SUCCESS] KL decomposition saved: kl_decomposition.json
```

5 Mathematical Foundations

5.1 The Polarity Index Method (PIM)

The Polarity Index Method transforms amino acid sequences into 16-dimensional vectors representing polarity transition frequencies.

Definition 5.1 (Amino Acid Polarity)**.** *Let $\mathcal{A} = \{A_1, \ldots, A_{20}\}$ be the set of the 20 standard amino acids. We define the polarity function $\phi : \mathcal{A} \to \{P^+, P^-, N, NP\}$ as:*

$$\phi(A) = \begin{cases} P^+ & \text{if } A \in \{H, K, R\} \\ P^- & \text{if } A \in \{D, E\} \\ N & \text{if } A \in \{C, G, N, Q, S, T, Y\} \\ NP & \text{if } A \in \{A, F, I, L, M, P, V, W\} \end{cases} \tag{1}$$

Definition 5.2 (PIM Vector)**.** *For a protein sequence $S = s_1 s_2 \ldots s_L$ of length L, we define the PIM vector $v \in \mathbb{R}^{16}$ as:*

$$v_i = \frac{1}{L-1} \sum_{k=1}^{L-1} \mathbf{1}\{\phi(s_k) = a, \phi(s_{k+1}) = b\} \tag{2}$$

where i is the index corresponding to the transition $a \to b$ in the order of the 16 possible transitions.

5.2 Wedge Product and Corrected Similarity

The wedge product captures the geometric relationship between two PIM vectors.

Definition 5.3 (Truncated Wedge Product)**.** *Let $v, w \in \mathbb{R}^{16}$ be two PIM vectors. We define the truncated wedge product over the set of biologically relevant pairs $\mathcal{K}$:*

$$\mathcal{K} = \{(0,5), (1,4), (2,6), (3,7), (10,11), (14,15)\} \tag{3}$$

The truncated wedge product is:

$$\omega_{\mathcal{K}}(v,w)_k = v_{i_k} w_{j_k} - v_{j_k} w_{i_k} \quad \textit{for } (i_k, j_k) \in \mathcal{K} \tag{4}$$

Definition 5.4 (Corrected Wedge Similarity)**.** *The corrected wedge similarity between two PIM vectors $v, w \in \mathbb{R}^{16}$ is:*

$$\text{WedgeSim}(v,w) = 1 - \frac{\|\omega_{\mathcal{K}}(v,w)\|_2}{\|v\|_2 \|w\|_2} \tag{5}$$

This similarity satisfies $0 \leq \text{WedgeSim}(v,w) \leq 1$, where $\text{WedgeSim}(v,w) = 1$ indicates identical vectors and $\text{WedgeSim}(v,w) = 0$ indicates orthogonal vectors.

5.3 Specular Reflection

Specular reflection captures the idea of charge-inverted functional equivalence.

Definition 5.5 (Reflection Normal Vector)**.** *The normal vector $n \in \mathbb{R}^{16}$ for specular reflection is:*

$$n_i = \begin{cases} +1 & \textit{if } i \in \{0,1,4,5\} \textit{ (charge components)} \\ 0 & \textit{otherwise} \end{cases} \tag{6}$$

normalized as $n = n/\|n\|_2$.

Definition 5.6 (Specular Reflection)**.** *The specular reflection of a PIM vector $v \in \mathbb{R}^{16}$ is:*

$$\text{SpecRef}(v) = v - 2(v \cdot n)n \tag{7}$$

This transformation inverts the sign of the charge components, simulating a charge inversion.

5.4 Clifford Signature

The Clifford signature provides a comprehensive characterization of a protein's polarity profile.

Definition 5.7 (Clifford Signature)**.** *The Clifford signature of a PIM vector $v \in \mathbb{R}^{16}$ is:*

$$\Sigma(v) = \big(\|v\|,\ \mathrm{AutoRef}(v),\ \mathrm{HydroProj}(v),\ \mathrm{ChargeProj}(v),\ \mathrm{AutoRot}(v),\ H(v),\ G(v),\ C(v),\ M(v),\ I(v)\big) \tag{8}$$

where:

$$\mathrm{AutoRef}(v) = \mathrm{WedgeSim}(v, \mathrm{SpecRef}(v)) \tag{9}$$

$$\mathrm{HydroProj}(v) = |v_{10} + v_{15}| \tag{10}$$

$$\mathrm{ChargeProj}(v) = |v_0 + v_5| \tag{11}$$

$$\mathrm{AutoRot}(v) = \mathrm{WedgeSim}(v, \mathit{roll}(v, 4)) \tag{12}$$

5.5 Shannon Entropy

The Shannon entropy measures the diversity of the polarity distribution.

Definition 5.8 (Shannon Entropy)**.** *For a PIM vector $v \in \mathbb{R}^{16}$, the Shannon entropy is:*

$$H(v) = -\sum_{i=1}^{16} p_i \log_2 p_i \tag{13}$$

where $p_i = |v_i| / \sum_{j=1}^{16} |v_j|$ and $p_i > 0$.

5.6 Jensen-Shannon Divergence

The Jensen-Shannon divergence measures the similarity between two probability distributions.

Definition 5.9 (Jensen-Shannon Divergence)**.** *For two PIM vectors $v, w \in \mathbb{R}^{16}$, the Jensen-Shannon divergence is:*

$$JS(v, w) = \frac{1}{2} KL(p \parallel m) + \frac{1}{2} KL(q \parallel m) \tag{14}$$

where $p = |v| / \sum |v|$, $q = |w| / \sum |w|$, $m = (p+q)/2$, and KL is the Kullback-Leibler divergence.

5.7 Gini Coefficient

The Gini coefficient measures the inequality in the polarity distribution.

Definition 5.10 (Gini Coefficient)**.** *For a PIM vector $v \in \mathbb{R}^{16}$, the Gini coefficient is:*

$$G(v) = \frac{\sum_{i=1}^{16} \sum_{j=1}^{16} |v_i - v_j|}{2n \sum_{i=1}^{16} v_i} \tag{15}$$

where $n = 16$ and $v_i \geq 0$.

5.8 Hellinger Distance

The Hellinger distance measures the similarity between two probability distributions.

Definition 5.11 (Hellinger Distance)**.** *For two PIM vectors $v, w \in \mathbb{R}^{16}$, the Hellinger distance is:*

$$H_d(v, w) = \sqrt{1 - \sum_{i=1}^{16} \sqrt{p_i q_i}} \tag{16}$$

where $p_i = |v_i| / \sum |v|$ and $q_i = |w_i| / \sum |w|$.

5.9 Spearman Correlation

The Spearman correlation measures the monotonic relationship between two PIM vectors.

Definition 5.12 (Spearman Correlation)**.** *For two PIM vectors $v, w \in \mathbb{R}^{16}$, the Spearman correlation is:*

$$\rho(v, w) = \frac{\sum_{i=1}^{16} (R_i - \bar{R})(S_i - \bar{S})}{\sqrt{\sum_{i=1}^{16} (R_i - \bar{R})^2 \sum_{i=1}^{16} (S_i - \bar{S})^2}} \tag{17}$$

where R_i and S_i are the ranks of v_i and w_i, respectively.

5.10 Moran's I

Moran's I measures spatial autocorrelation in the polarity profile.

Definition 5.13 (Moran's I)**.** *For a PIM vector $v \in \mathbb{R}^{16}$ with weight matrix W, Moran's I is:*

$$I(v) = \frac{n}{\sum_{i=1}^{16} \sum_{j=1}^{16} W_{ij}} \cdot \frac{\sum_{i=1}^{16} \sum_{j=1}^{16} W_{ij} (v_i - \bar{v})(v_j - \bar{v})}{\sum_{i=1}^{16} (v_i - \bar{v})^2} \tag{18}$$

where $\bar{v}$ is the mean of v.

5.11 Fractal Dimension

The fractal dimension measures the self-similarity of the polarity profile.

Definition 5.14 (Fractal Dimension). *For a PIM vector $v \in \mathbb{R}^{16}$, the fractal dimension is:*

$$D_f(v) = \lim_{\epsilon \to 0} \frac{\log N(\epsilon)}{\log(1/\epsilon)} \tag{19}$$

where $N(\epsilon)$ is the number of boxes of size ϵ needed to cover the profile.

5.12 Radon Transform

The Radon transform projects the polarity profile onto lines at various angles.

Definition 5.15 (Radon Transform). *For a PIM vector $v \in \mathbb{R}^{16}$, the Radon transform is:*

$$\mathcal{R}v(\theta, \rho) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\delta(\rho - x\cos\theta - y\sin\theta)\, dx\, dy \tag{20}$$

where $f(x, y)$ is the 2D representation of v.

5.13 Wasserstein Distance

The Wasserstein distance measures the cost of transforming one distribution into another.

Definition 5.16 (Wasserstein Distance). *For two PIM vectors $v, w \in \mathbb{R}^{16}$, the Wasserstein-1 distance is:*

$$W_1(v, w) = \sum_{i=1}^{16} |CDF_v(i) - CDF_w(i)| \tag{21}$$

where CDF is the cumulative distribution function.

5.14 Karhunen-Loève Decomposition

The Karhunen-Loève decomposition identifies the most informative components of the polarity profile.

Definition 5.17 (Karhunen-Loève Decomposition). *Let $\{v_1, \ldots, v_n\}$ be a set of PIM vectors. The covariance matrix is:*

$$C = \frac{1}{n-1} \sum_{i=1}^{n} (v_i - \bar{v})(v_i - \bar{v})^T \tag{22}$$

The Karhunen-Loève decomposition solves:

$$C\phi_k = \lambda_k \phi_k, \quad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{16} \geq 0 \tag{23}$$

The explained variance for component k is:

$$\mathrm{EV}_k = \frac{\lambda_k}{\sum_{j=1}^{16} \lambda_j} \tag{24}$$

5.15 Hodge Dual

The Hodge dual maps a k-vector to an (n-k)-vector.

Definition 5.18 (Hodge Dual). *For a vector $v \in \mathbb{R}^{16}$, the Hodge dual with pseudoscalar I is:*

$$\star(v) = v \cdot I^{-1} \tag{25}$$

where $I = e_1 \wedge e_2 \wedge \cdots \wedge e_{16}$.

5.16 Grassmann Distance

The Grassmann distance measures the distance between two subspaces.

Definition 5.19 (Grassmann Projection Distance). *For two PIM vectors $v, w \in \mathbb{R}^{16}$, the Grassmann projection distance is:*

$$d_G(v, w) = \frac{1}{\sqrt{2}}\|P_v - P_w\|_F \tag{26}$$

where $P_v = vv^T/\|v\|^2$ and $P_w = ww^T/\|w\|^2$.

5.17 Fubini-Study Metric

The Fubini-Study metric measures the distance in projective space.

Definition 5.20 (Fubini-Study Metric). *For two PIM vectors $v, w \in \mathbb{R}^{16}$, the Fubini-Study distance is:*

$$d_{FS}(v, w) = \arccos\left(\frac{|v \cdot w|}{\|v\|\|w\|}\right) \tag{27}$$

5.18 Ricci Curvature

The Ricci curvature measures the curvature of the Grassmann manifold.

Definition 5.21 (Ricci Curvature Tensor). *For a PIM vector $v \in \mathbb{R}^{16}$, the Ricci curvature tensor is:*

$$Ric(v) = \frac{n-1}{2}I + \frac{1}{2}(I - P_v) \tag{28}$$

where $P_v = vv^T/\|v\|^2$ and $n = 16$.

6 LUJO Analysis Results

6.1 Processing Statistics

The LUJO run processed the following data:

Table 4: Processing statistics

Metric	**Value**
Total sequences read	151,066,931
Valid PIMs	151,066,930
Validity rate	100.00%
Data processed	72.17 GB
Processing time	13 h 21 min
Average speed	3,141 seq/s
Peak memory usage	500 MB
Number of groups	26

6.2 Similarity to Other Groups

The wedge similarity between LUJO and other groups reveals the closest functional relationships:

Table 5: Top groups similar to LUJO

Group	**Wedge Similarity**
CPP (Cell-Penetrating Peptides)	0.1982
NON_CPP	0.1360
NILE1 (West Nile Virus)	0.1709
LASV (Lassa Virus)	0.0881
PARTIALLY_FOLDED	0.0438
REVIEWED_HUMAN	0.0247
UNREVIEWED_ALL	0.0204

6.3 Chemical Profile of LUJO

The complete chemical profile of LUJO includes 54 properties:

6.4 Intrinsic Disorder Prediction (PIDP)

The PIDP analysis provides complementary structural information:

6.5 Designed Therapeutic Peptide

The therapeutic peptide design module generated a competitor peptide:

6.6 Comparison with Known Inhibitors

The designed peptide compares favorably with known inhibitors:

Table 6: Selected chemical properties of LUJO

Property	**Value**
Net charge (pH 7.4)	+0.052
Charge density	0.085
Charge balance	0.808
pI	5.97
GRAVY	0.253
Hydrophobicity score	0.196
N-glycosylation sites	12
O-glycosylation sites	64
Phosphorylation sites	32
Acetylation sites	7
Solubility (mg/mL)	12.81
ΔG (kcal/mol)	-6.76
T_m (°C)	41.04
Stability score	0.552
Calcium binding	0.303
Zinc binding	0.450
Membrane affinity	0.831
Best lipid	Cholesterol
Optimal pH	6.97
Salt tolerance (mM)	241.50

6.7 Karhunen-Loève Decomposition

The KL decomposition reveals the low-dimensional structure:

6.8 Peptide Recommendations

Based on the therapeutic profile, the following recommendations are provided:

1. **Synthesis**: Sequence NDLKNAAAAAA by solid-phase synthesis.
2. **Formulation**: PBS pH 7.4 buffer.
3. **Protection**: Add protecting groups at N and Q (avoid deamidation).
4. **Validation**: GP binding assays (SPR/ITC).
5. **Optimization**: Mutate critical residues to improve activity.

Table 7: PIDP results for LUJO

Tool	Threshold	Disorder (%)
metapredict	0.3	12.33
metapredict	0.4	9.91
metapredict	0.5	8.15
metapredict (mean)	—	0.1655
metapredict (max)	—	1.0
metapredict (min)	—	0.02
metapredict (std)	—	0.2248
AIUPred	0.3	20.26
AIUPred	0.4	12.78
AIUPred	0.5	9.69
AIUPred (mean)	—	0.1919
AIUPred (max)	—	0.8846
AIUPred (min)	—	0.0317
AIUPred (std)	—	0.1894
AIUPred (redox sensitivity)	—	0.029

Table 8: Designed competitor peptide

Property	Value
Sequence	NDLKNAAAAAA
Length	11 aa
Net charge	0.0
Molecular weight	1209.3 Da
Hydrophobicity	0.018
pI	6.0
Solubility	14.96 mg/mL
Predicted antiviral activity	0.500

7 Interpretation of Results

7.1 Intrinsic Disorder Interpretation

The low disorder of LUJO (8.15–9.69% at threshold 0.5) confirms it is a predominantly ordered protein, consistent with its role as a structural glycoprotein involved in receptor binding and membrane fusion.

Practical interpretation:

- <10% disorder: Ordered protein, suitable for crystallization.
- 10–30% disorder: Flexible regions important for function.
- >30% disorder: Mostly disordered protein (IDP).

Table 9: Affinity comparison with inhibitors

Inhibitor	IC50 (nM)	Type
Designed peptide	0.012	Competitor
mAb114	0.009	Antibody
REGN-EB3	0.012	Antibody
ZMapp	0.015	Antibody
Remdesivir	0.080	Small molecule
Favipiravir	0.250	Small molecule

Table 10: KL decomposition (first 8 components)

Component	Eigenvalue	Explained Variance (%)
1	0.012775	32.49
2	0.009003	22.90
3	0.004103	10.44
4	0.003444	8.76
5	0.002596	6.60
6	0.001883	4.79
7	0.001481	3.77
8	0.001062	2.70
Cumulative		92.45

7.2 Chemical Properties Interpretation

- **Positive net charge (+0.052)**: Slightly basic protein at physiological pH.
- **pI 5.97**: Slightly acidic isoelectric point.
- **GRAVY 0.253**: Slightly hydrophobic.
- **Glycosylation sites**: 12 N-linked and 64 O-linked sites suggest high glycosylation, important for immune evasion.
- **Stability**: $\Delta G = -6.76$ kcal/mol indicates a stable protein.

7.3 Drug-Binding Hot Spots

Three drug-binding hot spots were identified:

1. **Hydrophobic patch (fusion loop-like)**: Score 0.339, residues D512, E515, H516, I518, W519, A520.
2. **Mixed polar/hydrophobic region (binding loop-like)**: Score 0.268, residues F134, F136, Y139, D142, K144.
3. **Charge-rich region (CR2-like)**: Score 0.104, residues H54, E64, D66, K68, H71.

8 Troubleshooting

8.1 Common Issues and Solutions

Table 11: Common issues and solutions

Issue	Solution
File not found	Verify that the .dat0 file exists
MemoryError	Reduce MAX_STORED_PROTEINS_PER_GROUP
No valid PIM vectors	Check sequence length (> 2 aa)
Slow execution	Reduce BATCH_SIZE or use fewer groups
Import error	Install missing package with pip
PIDP not available	Install metapredict/aiupred or disable
Singular covariance matrix	Check for duplicate sequences in group

8.2 Performance Optimization

For large datasets, consider these optimizations:

Table 12: Performance optimization strategies

Strategy	How to implement
Reduce sample size	Set MAX_STORED_PROTEINS_PER_GROUP = 1000
Disable plots	Set GENERATE_PLOTS = False
Disable PIDP	Set USE_PIDP = False
Reduce batch size	Set BATCH_SIZE = 50000

9 Quick Reference

9.1 Main Commands

```
# Run full analysis
python3 SGPMAIN6.2v.py

# View results
ls results_v17_*/

# View chemical profile
cat results_v17_*/chemical_profile_lujo.csv

# View comparison
cat results_v17_*/comparison_lujo_vs_all.csv
```

```

# View PIDP results
cat results_v17_*/pidp_summary_all_targets.csv

# View KL decomposition
cat results_v17_*/kl_decomposition.json

# View therapeutic profile
cat results_v17_*/therapeutic_profile.json
```

Listing 2: Quick commands

9.2 File Locations

Table 13: File locations

File Type	Location
Input files	Current working directory
Output files	results_v17_YYYYMMDD_HHMMSS/
Configuration	SGPMAIN6.2v.py (edit directly)

9.3 Important Notes

1. Always backup input files before running.
2. Results are timestamped to prevent overwriting.
3. Large datasets may require several hours to process.
4. The program supports Ctrl+C for safe interruption.
5. Logs are saved for debugging and reproducibility.
6. PIDP requires metapredict and/or aiupred to be installed.
7. Complete source code available at: `https://github.com/polancodf2024/lujv-sgp-analysis`

10 References

1. Polanco, C., et al. (2015). Polarity Index Method for protein analysis. *Journal of Computational Biology*.
2. Hestenes, D. (1984). *Clifford Algebra to Geometric Calculus*. Springer.
3. Hestenes, D. (2017). *Geometric Algebra for Physicists*. Cambridge.
4. Dorst, L., et al. (2009). *Geometric Algebra for Computer Science*. Elsevier.

5. Edelman, A., et al. (1998). The geometry of algorithms with orthogonality constraints. *SIAM Journal on Matrix Analysis*.

6. metapredict: `https://github.com/idptools/metapredict`

7. AIUPred: `https://github.com/idptools/aiupred`